

A Variant Of Chaitin's Omega function

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Joint work with Yuxuan Li, Shuheng Zhang and Xuanheng Zhao

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Preliminary

Cantor space

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The **topology** on 2^ω is generated by $[\sigma]$ for all strings σ . These sets are clopen. The **Lebesgue measure** on 2^ω is defined by $\mu([\sigma]) = 2^{-|\sigma|}$.

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The Cantor space with this topology and measure can be identified with the closed interval $[0, 1] \subseteq \mathbb{R}$ equipped with its usual topology and measure, up to ignoring whatever happens at the dyadic rationals.

Kolmogorov complexity

Let M be a Turing machine with string input and output. The M -**complexity** of a string σ is the length of the shortest description of σ in M , i.e.

$$K_M(\sigma) = \min\{|\tau| : M(\tau) \downarrow = \sigma\}.$$

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A prefix-free machine U is **optimal** if for any prefix-free machine M there is a constant c such that $K_U(\sigma) \leq K_M(\sigma) + c$. For instance, $U(0^e 1 \tau) = M_e(\tau)$ where M_e is the e th prefix-free machine.

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We fix an optimal prefix-free machine U and let the **Kolmogorov complexity** of σ be $K(\sigma) = K_U(\sigma)$.

Randomness

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Note that the property of being random does not depend on the optimal machine in context.

Chaitin's Omega

The **Chaitin's constant**, often denoted by Ω , is defined to be the halting probability of a random program / input:

$$\Omega = \Omega_U = \mu(\text{dom}(U)) = \sum_{U(\tau) \downarrow} 2^{-|\tau|}$$

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Here are some well-known facts about Ω :

- Ω is left-c.e. (that is, $\{q \in \mathbb{Q} : q < \Omega\}$ is c.e.);
- Ω is 1-random;
- $\Omega \equiv_T \emptyset'$.

Kolmogorov weight

The **Kolmogorov weight** of a string σ is $2^{-K(\sigma)}$. Intuitively this is the probability that a random program with random input outputs σ , up to a multiplicative constant. It can be used as an alternative definition for Ω using

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Note that $\Omega_U \neq \hat{\Omega}_U$, but they are equivalent in the following sense:

Theorem (Kučera, Slaman, 2001; ...)

The following are equivalent for a real a :

- ① a is left-c.e. and 1-random;
- ② a is left-c.e. and $a \equiv_T \emptyset'$;
- ③ $a = \Omega_U$ for some optimal prefix-free machine U ;
- ④ $a = \hat{\Omega}_U$ for some optimal prefix-free machine U .

We call such reals **Omega numbers**.

Omega as a function

Literature

Recall $\Omega = \mu(\text{dom}(U)) = \sum_{U(\tau) \downarrow} 2^{-|\tau|}$ and $\hat{\Omega} = \sum_{\sigma} 2^{-K(\sigma)}$.

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Definition (Downey, Hirschfeldt, Miller, Nies, 2005)

Consider $x \mapsto \Omega^x = \mu(\text{dom}(U^x)) = \sum_{U^x(\tau) \downarrow} 2^{-|\tau|}$ where U is an optimal prefix-free oracle machine. This function is, e.g. not continuous.

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Definition (Hölzl, Merkle, Miller, Stephan, Yu, 2020)

Consider $x \mapsto \sum_{\sigma \prec x} 2^{-K(\sigma)}$. This function is, e.g. always continuous; and differentiable at x if and only if x is 1-random.

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Definition (Becher, Figueira, Grigorieff, Miller, 2006)

Consider $X \mapsto \Omega[X] = \sum_{U(\tau) \downarrow \in X} 2^{-|\tau|}$, where X is a set of strings.

Our variant of Omega function

We study the function

$$f(x) = \sum_{\sigma \leq_L x} 2^{-K(\sigma)}$$

and its image $f(2^\omega)$, where

- $\sigma <_L x$ if $\sigma(n) = 0$ while $x(n) = 1$ for some $n < |\sigma|$;
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Some quick observations:

- f is a function from 2^ω to $[0, 1]$;
- $f(1^\omega) = \sum_{\sigma} 2^{-K(\sigma)} = \hat{\Omega}$;
- $f(0^\omega) = \sum_n 2^{-K(0^n)}$ which is also an Omega number;
- f is strictly increasing and continuous;
- $f(10^\omega) = f(01^\omega) + \sum_n 2^{-K(10^n)}$.

Analytic properties

Differentiability - Density randomness

For a measurable $P \subseteq [0, 1]$, the **lower Lebesgue density** of a point $z \in (0, 1)$ in P is

$$\liminf_{\gamma, \delta \rightarrow 0} \frac{\mu([z - \gamma, z + \delta] \cap P)}{\gamma + \delta}.$$

A real number $z \in (0, 1)$ is a **density-one point** if it has lower Lebesgue density 1 in every effectively closed set P (that is, $P = [0, 1] - \bigcup_i (a_i, b_i)$ where a_i, b_i are rationals such that $i \mapsto a_i$ and $i \mapsto b_i$ are computable) containing z .

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Warning: We can mimic the above definition in 2^ω . A real number is called a **dyadic density-one point** if its binary expansion is a density-one point in 2^ω . Being dyadic density-one is strictly weaker than being density-one.

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Theorem (Khan, 2016)

*A 1-random real number x is a density-one point if and only if it is a dyadic density-one point. Such reals are called **density random**.*

Differentiability

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$$f(x) = \sum_{\sigma \preceq_L x} 2^{-K(\sigma)}.$$

Differentiability

Recall that

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That f is differentiable at density random points is a (rather) direct consequence of the following.

Definition (Bienvenu, Greenberg, Kučera, Nies, Turetsky, 2016)

A non-decreasing function $g : [0, 1] \rightarrow [0, 1]$ is **interval c.e.** if $g(0) = 0$ and $g(q) - g(p)$ is left-c.e. uniformly in all rational pairs $p < q$.

Definition (Miyabe, Nies, Zhang, 2016)

A real number x is density-random if and only if g is differentiable at x for each interval-c.e. function g .

Differentiability

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Theorem

f is differentiable at x if and only if x is density random.

Proof Sketch.

If x is density random, then f is differentiable at x since $f - f(0)$ is interval c.e.. Suppose that f is differentiable at x . Then do the following steps:

- Prove that x is 1-random. This is similar to proving that if $x \mapsto \sum_{\sigma \prec x} 2^{-K(\sigma)}$ is differentiable at x then x is 1-random.
- Prove that $f'(x) = 0$. This comes from the high initial segment complexity of 1-random x and some calculation.
- Check that x is a density-one point. This requires some more calculation.



Image as a generalized Cantor set

Recall that

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For each string σ , $f(\sigma 10^\omega) = f(\sigma 01^\omega) + \sum_n 2^{-K(\sigma 10^n)}$. There is no real x between $\sigma 10^\omega$ and $\sigma 01^\omega$ in the Cantor space, therefore this produces the gap in the image as a generalized Cantor set.

Image as a generalized Cantor set

Theorem

The image $f(2^\omega)$ is a null, nowhere dense, perfect set with Hausdauff dimension 1.

Proof Sketch.

Being nowhere dense and perfect follows from being a generalized Cantor set. For being null, observe that each string containing some 1 is of the form $\sigma 10^n$, therefore its Kolmogorov weight is contributed to the gap created by σ . Otherwise it is of the form 0^n and its Kolmogorov weight is contributed to $f(0^\omega)$. Therefore

$$f(1^\omega) = \hat{\Omega} = f(0^\omega) + \sum_{\text{all gaps}} \mu(\text{the gap})$$

so $\mu(f(2^\omega)) = 0$.

For Hausdauff dimension 1, the idea is that the “Cantor middle $1/n$ set” has Hausdauff dimension $\log 2 / (\log 2 + \log n - \log(n-1))$, and that the ratio of the gap removed at each level goes to 0 since K grows very slowly. □

Computability properties

Comparison with $x \mapsto \Omega^x$

Recall the classical Omega operator $x \mapsto \Omega^x = \mu(\text{dom}(U^x))$. By relativizing the aforementioned properties of Ω , we have

- ① Ω^x is left-c.e. relative to x ;
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For item (1), the corresponding property for f follows directly from that f is strictly increasing and that f uniformly maps rationals to left-c.e. reals. In particular, $f(x)$ is left-c.e. relative to x and x is right-c.e. relative to $f(x)$.

How random is $f(x)$?

Recall that

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Item (2) says that Ω^x is always 1-random relative to x , so in particular, Ω^x is always 1-random; and it can be arbitrarily random.

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- ① If x is left-c.e., then $f(x)$ is 1-random (in fact, an Omega number);
- ② $f(x)$ cannot be arbitrarily random: Observe that $f(2^\omega)$ is a $\Pi_1^0(\emptyset')$ class (a class of reals effectively closed relative to $\emptyset'$) and has measure 0, so $f(x)$ is not weakly 1-random relative to $\emptyset'$.

A real x is weakly 1-random relative to z if x is not in any $\Pi_1^0(z)$ class of measure 0. 1-randomness implies weakly 1-randomness.

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In particular, $f(x)$ can never be 2-random (that is, 1-random relative to $\emptyset'$).

How much computational power does $f(x)$ require/carry?

Recall that

$$f(x) = \sum_{\sigma \leq_L x} 2^{-K(\sigma)}.$$

Item (3) says that $\Omega^x \oplus x \equiv_T x'$, so Ω^x combined with x can always compute something that x cannot compute.

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- ① $x' \geq_T x \oplus \emptyset' \geq_T f(x)$ (uniformly);
- ② $f(x)' \geq_T f(x) \oplus \emptyset' \geq_T x$ (uniformly).

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Although we cannot expect to get x' from $f(x) \oplus x$, we can at least always get $\emptyset'$:

- ③ $f(x) \oplus x \geq_T \emptyset'$.

Randomness properties

When is $f(x)$ random?

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Theorem (Miller, 2009)

A real x is weakly low for K if and only if x is low for Ω .

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Theorem

$f(x)$ is 1-random relative to x if and only if x is weakly low for K (if and only if x is low for Ω).

When is $f(x)$ random?

Corollary

$f(x)$ is 1-random for almost all x .

Proof.

Almost all x are 2-random (that is, 1-random relative to $\emptyset'$). Such x are weakly low for K . Thus $f(x)$ is 1-random relative to x , in particular 1-random for such x . □

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Theorem

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Proof.

Recall that each string σ creates a gap $(f(\sigma 01^\omega), f(\sigma 10^\omega))$ in the image $f(2^\omega)$. The size of this gap is $\sum_n 2^{-K(\sigma 10^n)}$.

Adjacent gaps are of comparable size, that is, the size of a gap created by σ is approximately equal to the size of the gap created by $\sigma * 0$ and $\sigma * 1$. This is because $K(\sigma 10^n) \approx K(\sigma 100^n) \approx K(\sigma 110^n)$.

Then see the whiteboard.



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Proof.

We start with $x_0 = 0^\omega$. In stage s , if the $s(c+1)$ th bit to the $(s+1)(c+1)$ th bit of $f(x_s)$ are precisely 10^c , we apply the Lemma with $n = s(c+1) + 1$ to get $x_{s+1} = y$, so $f(x_{s+1})$ changes from $f(x_s)$ by at least $2^{-s(c+1)-c-1}$ and by at most $2^{-s(c+1)-1}$. Then

- Bits before the $s(c+1)$ th bit are unchanged;
- At least one bit from the $s(c+1)$ th bit to the $(s+1)(c+1)$ th bit changes, so it is no more 10^c ;
- Bits after the $(s+1)(c+1)$ th bit might change as they want, but we don't care as we'll go fix them in later stages.

This way we get a sequence of $f(x_s)$ converging to some $f(x)$. Since f is continuous, x_s converges to x . Now 10^c does not appear as the $s(c+1)$ th bit to the $(s+1)(c+1)$ th bit of $f(x)$, so $f(x)$ is not normal. □

When is $f(x)$ non-random?

Theorem

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By altering details of the above proof, we get:

Corollary

There is $x \leq_T \emptyset'$ such that $f(x)$ is not normal.

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Corollary

There are continuum many x such that $f(x)$ is not normal.

Weight and halting probability

Transferring to halting probability

Recall that the following sums

$$\Omega = \mu(\text{dom}(U)) = \sum_{U(\tau) \downarrow} 2^{-|\tau|},$$

$$\hat{\Omega} = f(1^\omega) = \sum_{\sigma} 2^{-K(\sigma)}$$

can both be definitions of Omega and share similar properties; while we defined our function function

$$f(x) = \sum_{\sigma \leq_L x} 2^{-K(\sigma)}.$$

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Thus we can also study

$$g(x) = \sum_{U(\tau) \downarrow \leq_L x} 2^{-|\tau|},$$

so that $\Omega = g(1^\omega)$.

Transferring to halting probability

Recall that

$$f_U(x) = \sum_{\sigma \leq_L x} 2^{-K_U(\sigma)},$$

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For each optimal prefix-free machine U , there is an optimal prefix-free machine V such that $f_U = g_V$.

For each optimal prefix-free machine V and left-c.e. real x , there is an optimal prefix-free machine U such that $f_U(x) = g_V(x)$.

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In particular, machine-independent properties for g also hold for f . Whether the converse hold (instead of only on a single left-c.e. real x) or not is more obscure and is left open.

Transferring theorems

Recall that

$$g(x) = \sum_{U(\tau) \downarrow \leq_L x} 2^{-|\tau|}.$$

Also recall that the operator $X \mapsto \Omega[X] = \sum_{U(\tau) \downarrow \in X} 2^{-|\tau|}$ is already studied by Becher et al.; and note that $g(x) = \Omega[\{\sigma : \sigma \leq_L x\}]$. Therefore we can further study the properties of f by applying known results of $X \mapsto \Omega[X]$.

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Theorem (Becher, Figueira, Grigorieff, Miller, 2006)

If $n \geq 2$ and X is Σ_n^0 -complete (that is, X is Σ_n^0 and $X \equiv_T \emptyset^{(n)}$) or Π_n^0 -complete, then $\Omega[X]$ is 1-random.

Corollary

If $n \geq 2$ and x is left- Σ_n^0 -complete (that is, x is left-c.e. relative to $\emptyset^{(n-1)}$ and $x \equiv_T \emptyset^{(n)}$) or right- Π_n^0 -complete, then $f(x)$ is 1-random.

Generalizing theorems

We can also generalize “existence” theorem about the operator $X \mapsto \Omega[X] = \sum_{U(\tau) \downarrow \in X} 2^{-|\tau|}$ to the setting of our function f .

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Theorem (Becher, Figueira, Grigorieff, Miller, 2006)

There is an infinite $X \leq_T \emptyset'$ such that $\Omega[X]$ is not normal.

This can be regarded as a corollary of our result mentioned before:

Theorem

There is $x \leq_T \emptyset'$ such that $f(x)$ is not normal.

Equivalent characterization of Omega number

The operator $X \mapsto \Omega[X] = \sum_{U(\tau) \downarrow \in X} 2^{-|\tau|}$ also provides a nice characterization of Omega numbers. Recall that an Omega number is a left-c.e. complete 1-random real, and these are precisely all halting probabilities of optimal prefix-free machines.

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Theorem (Becher, Figueira, Grigorieff, Miller, 2006)

The following are equivalent for a real a :

- ① *a is an Omega number;*
- ② *There is a c.e. set X and an optimal prefix-free machine V such that $a = \Omega_V[X]$.*
- ③ *For each non-empty c.e. set X there is an optimal prefix-free machine V such that $a = \Omega_V[X]$.*

Equivalent characterization of Omega number

We have a similar characterization in our setting:

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- ③ *For each left-c.e. real x there is an optimal prefix-free machine V such that $a = f_V(x)$.*

Remark: It is not immediate obvious that the V constructed in these theorems can be made universal (that is, for each prefix-free machine M , there is a string ρ_M such that $V(\rho_M\tau) = M(\tau)$ for all τ), instead of only optimal.

Left-c.e. randomness

Range structure

Recall that

$$f(x) = \sum_{\sigma \leq_L x} 2^{-K(\sigma)},$$

and we have mentioned that $f(2^\omega)$ is $\Pi_1^0(\emptyset')$. To be more precise, this is because

$$f(2^\omega) = [0, 1] - [0, f(0^\infty)) - (f(1^\infty), 1] - \bigcup_{\sigma} \text{The gap created by } \sigma$$

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E.g. The endpoints are left-c.e. instead of only $\emptyset'$ -computable. The remaining measure goes to 0 effectively, in contrast the remaining measure of some arbitrary $\Pi_1^0(\emptyset')$ class only goes to 0 effectively in $\emptyset'$.

Left-c.e. test and left-c.e. randomness

A **Martin-Löf test** is a sequence $\{V_n\}_{n \in \omega} = \{\bigcup_i [a_i^n, b_i^n]\}_{n \in \omega}$ such that a_i^n and b_i^n are computable uniformly in n and i , and $\mu(V_n) \leq 2^{-n}$. A real x fails the test if $x \in \bigcap_n V_n$. A real x is **Martin-Löf random** if it passes all Martin-Löf tests. It is well-known that 1-randomness and Martin-Löf randomness are equivalent.

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Immediate observation: $f(2^\omega)$ is a left-c.e. test, therefore any $f(x)$ is not left-c.e. random.

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Theorem

Left-c.e. randomness is equivalent to 2-randomness (1-randomness relative to $\emptyset'$).

Finitely left-c.e. test and finitely left-c.e. randomness

What if we further refine?

Definition

A **finitely left-c.e. test** is a sequence $\{V_n\}_{n \in \omega} = \{\bigcup_i [a_i^n, b_i^n]\}_{n \in \omega}$ such that a_i^n and b_i^n are left-c.e. uniformly in n and i , for each n all but finitely many b_i^n are 0, and $\mu(V_n) \leq 2^{-n}$. A real x fails the test if $x \in \bigcap_n V_n$. A real x is **finitely left-c.e. random** if it passes all finitely left-c.e. tests.

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(Not so immediate) Observation: $f(2^\omega)$ is a finitely left-c.e. test, therefore any $f(x)$ is not finitely left-c.e. random.

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Remark: In general we can show that all finitely left-c.e. tests test out the same reals as all null generalized Cantor sets with uniformly left-c.e. endpoints.

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Remark: In general we can show that all finitely left-c.e. tests test out the same reals as all null generalized Cantor sets with uniformly left-c.e. endpoints.

Theorem

Finitely left-c.e. randomness is equivalent to weakly 1-randomness relative to $\emptyset'$.

Open problems

The main open problem left in the work is the following:

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Is there x such that $f(x)$ is rational / computable / right-c.e.?

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More generally we could ask:

Problem

Fix a rational / computable / right-c.e. y . Is there an optimal machine V such that $y \in f(2^\omega)$?

Thanks